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NetLogo: Complexity Made Simple

This article introduces NetLogo, a multi-turtle version of Logo.

The first time I came across NetLogo (<http://ccl.northwestern.edu/netlogo/>) several years ago, it fascinated me. I was amazed to see the seemingly intelligent behaviour of ants modelled by a fairly simple parameter. On the other hand, the poor traffic lights leading to traffic jams can also be modelled simply. The current version of NetLogo 5.01 comes with a large set of models you can explore. NetLogo, as its name suggests, is a multi-turtle version of Logo, which you may have used as a very young child! Yet, surprisingly, it can lead to a comprehension of seemingly intractable behaviour using rather simple models.

The very first model that I would recommend to get an idea of NetLogo is the creation of groups in a party. The group sizes are not uniform, and one may have experienced that a fairly large number of groups at a party tend to have members of the same sex—and this is not just in India. The model explores the idea that if the percentage of members of the opposite sex in a group is larger than some tolerance value, a person gets uncomfortable, and leaves the group. As the documentation on the NetLogo site states:

“... a surprising result of the Party model is that even if tolerance is relatively high, a great deal of separation between the sexes occurs.”

Well, the model may be incorrect; however, a more likely scenario is that we have never been exposed to reasoning about complexity. Today, our world is more interconnected than ever before. Interactions between entities often result in consequences that may not have been anticipated or understood. For example, I was struck by the protests from employees of telecom companies who are being laid off because of the 2G scam. If the companies involved lose a few billions, would it have the same impact on their owners as on the people losing their jobs? The owners of the companies, especially in the richer countries, may be financial institutions; however, the money could very well be from pension or mutual fund investments of the ‘common’ citizens! So, who is really being penalised?

How do we learn to reason about complexity and the surprising behaviour that may emerge out of interactions between entities? Computers have proved to be an indispensable tool for making sense of such phenomena, by enabling the simulation and visualisation of the evolution of complex systems, over time. NetLogo makes this capability accessible to all. See <http://ccl.northwestern.edu/papers/MEE/> for the background.

As mentioned earlier, a number of sample models for

various subjects are included. For example, for computer science, there is the model for the classic problem involving the dining philosophers. The simulation makes it easier to understand the issues involved in synchronising concurrent processes, and the prevention of deadlocks, by the introduction of cooperation. Each model comes with information about the model and how it works, including guidance about things to take note of. The code for the model is also included. The simulation, the information, and the code are very easily accessible as tabs in the user interface.

Another interesting model included is the page-rank model. The agent-based perspective gives a deeper understanding of the algorithm. For example, a page that has only one link, but comes from a highly ranked page, will also have a high rank. But how does one tell that a starting page is highly ranked? NetLogo provides two algorithms, which start with each page being equally ranked, but eventually converge to the same page rank. I was fascinated by the simulation of surfers that is used to arrive at the page’s ranking.

The models included in the distribution are grouped by various disciplines. Among the social sciences, learners may want to explore ‘Wealth Distribution’ or the familiar “The rich get richer and the poor get poorer” saying. Or one can explore ‘Altruism’ to study what conditions can make the altruistic genes beat the selfish genes. In the earth sciences, a model of climate change is expected, and should be fun to explore.

Last, but not least, there is the game section, in which one can play the classics – Pacman and Tetris. The fun part for technology enthusiasts would be to study the game code implemented as a network of turtles!

NetLogo models can run as Java applets in the browser. Try some at <http://ccl.northwestern.edu/netlogo/models/community/>.

Finally, you may wish to explore whether our species will survive—and the place to start would be the ‘Wolf-Sheep’ predator model in biology. If you are an entrepreneur, simulate what makes a social network tick, and create a predator app that eats the currently successful ones! 

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